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Paper Id: 238441 | Roll No. | | | | | | | | | | |

B TECH
(SEM IV) THEORY EXAMINATION 2022-23
ANALOG CIRCUITS

Time: 3 Hours | *Total Marks: 100*

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 10 = 20

(a) Draw V-I characteristics of ideal and piecewise linear model of diode with support of mathematical equations.

(b) Define knee voltage and breakdown voltage of a diode and write down it's mathematical expressions.

(c) Sketch the equivalent circuit model of the trans-resistance and write the ideal values of terminal characteristics.

(d) Why common emitter configuration of BJT preferred to design an amplifier circuit among other BJT configurations?

(e) Draw the circuit diagram of a unity gain amplifier using op-amp and write is applications.

(f) Define different parameters used in high frequency hybrid-π model.

(g) An amplifier has a mid-band gain of 120 and bandwidth of 250 KHz. If 6% negativefeedback is introduced, find the new bandwidth and gain.

(h) Define minimum sustainable voltage and maximum usable load of a Current mirrorCircuits.

(i) Write the ideal characteristics of an operational amplifier.

(j) Explain Barkhausen criterion.

SECTION B

2. Attempt any *three* of the following: 10x3=30

(a) Calculate current I_D and output voltage V_0 for the circuit shown in Fig.1

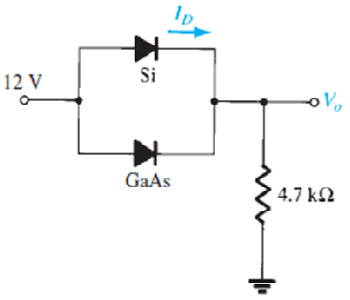


Fig.1

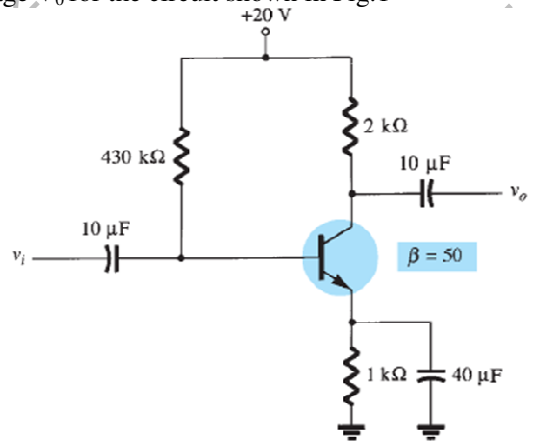


Fig.2

(b) Find the DC bias point (operating point) for the circuit shown in Fig.2

(c) Derive the expression of frequency of oscillation for RC phase shift oscillator. And narrate it's salient features.

(d) Describe the basic structure and operation of differential amplifier using BJT.

(e) Illustrate super diode and describe the operation of full wave precision rectifier with suitable circuit diagram.

SECTION C

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3. Attempt any *one* part of the following: 10x1=10

(a) Sketch the frequency response of single stage RC coupled BJT amplifier and describe the same for multistage amplifier.

(b) Derive the expression of voltage gain, input resistance and output resistance of common source amplifier.

4. Attempt any *one* part of the following: 10x1=10

(a) The BJT in the circuit of Fig.3 has $\beta = 100$.
(i) Find the DC collector current and the de voltage at thecollector.
(ii) Replacing the transistor by its T model, draw the smallsignalequivalent circuit of the amplifier. Analyze the resultingcircuit to determine the voltage gain v_0/v_i.

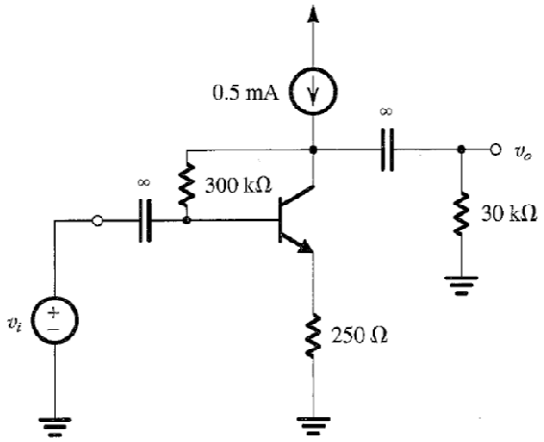


Fig.3

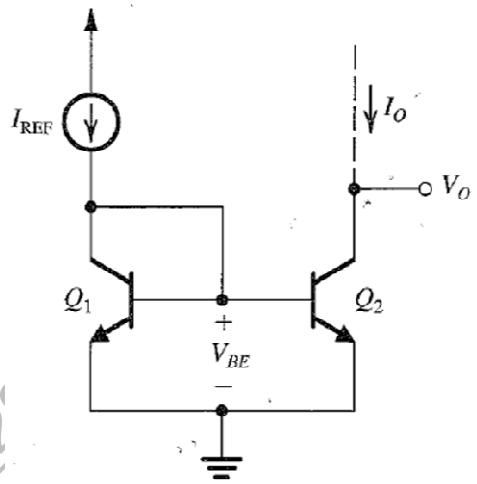


Fig.4

(b) Consider the basic BJT current mirror of Fig.4 forthe case in which Q_2 has m times the area of emitter base junction of Q1. Show thatthe current transfer ratio is given by$I_0 = m\ I_{REF}$.

5. Attempt any *one* part of the following: 10x1=10

(a) Illustrate the different design stages of operational amplifier.

(b) Design an astablemultivibrator circuit using op-amp having frequency of oscillation 5KHz.

6. Attempt any *one* part of the following: 10x1=10

(a) Discuss the operation of negative feedback with suitable block diagram and describe its properties in detail.

(b) Explain the circuit of wide bandpass active filter and design a wide bandpass filter having lower cut off frequency 3KHz and higher cutoff frequency 30KHz with passband gain of 12 dB.

7. Attempt any *one* part of the following: 10x1=10

(a) Explain the operation of class B amplifier and discuss its drawback. Any one way to overcome the drawback.

(b) Design the following circuits using op-amp
i)Unity gain amplifier
ii)Integrator circuit.

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